MIL-P-51458(EA)
26 June 1979
(See section 6)

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, VX-2, XM736 W/O

CANISTER: NM, XM28, LOADING,

ASSEMBLY AND PACKING

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembly and packing of one type of projectile.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification For.

FSC 1320

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 15-12-155 - Projectile, 8 Inch, VX-2, XM736 W/O Canister: NM, XM28

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing TDPL 15-12-155 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the projectile shall conform to the specifications and drawings listed on Drawing TDPL 15-12-155 and on the subsidiary drawings.

3.2 Assembly. The projectile shall be assembled as shown on Drawing E15-12-155 (see 6.2).

3.3 Ballistic functioning.

3.3.1 Canister. Both canister burst disks shall separate from the canisters. After separation of disks, any metal tab remaining of either disk shall not exceed 1 square inch in area.

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3.3.2 Projectile. The following requirements shall be met when tested as specified in 4.4.4:

a. The projectile shall launch and fly to the target area with no evidence of liquid leakage.

b. The projectile hardware shall remain intact until function of the time fuze.

c. Upon fuze functioning the canisters shall be expelled from the base of the projectile body.

This requirement shall also apply to those rounds fired at an adjusted pressure (15 percent overcharge), except that at this charge, expulsion of canisters shall not be considered. In all cases, canister expulsion failure due to fuze failure shall not be cause for lot rejection.

3.4 Preproduction. Prior to the start of regular production, a preproduction lot shall be produced in accordance with this specification for examination and tests.

3.5 Workmanship. The projectile and its components shall be free from foreign matter (oil, grease or dirt) and damage such as dents, cracks, burrs, sharp edges or deformation which could affect their intended use and the projectile shall comply with the requirements of materials, parts and assemblies conforming to MIL-A-2550.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A lot of 12 projectiles shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive test. The preproduction lot shall be examined and tested in accordance with the classification of defects.

4.3.2.2 For destructive test (ballistic functioning). The preproduction lot shall be tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction lot shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The acceptance number is zero. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot of projectiles shall be identified and controlled in accordance with MIL-STD-1168. However, no more than one interfix of each component (charges, metal parts assembly or bases) from one manufacturer shall be represented in any one lot of assembled projectiles.

Only one lot of XM27 canisters shall be permitted in any one lot of assembled projectiles.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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4.4.2.2 For destructive test (ballistic functioning). Six projectiles (one pallet) shall be taken at random from each lot.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample projectiles and the level A preparation for delivery shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive test. The sample projectiles shall be tested in accordance with 4.4.4.

4.4.3.3 Critical characteristics. Each projectile in the lot shall be inspected for its critical characteristic listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Projectile, 8 Inch, VX-2, XM736 w/o canister: NM, XM28 (Drawing E15-12-155).

Categories	Defects
Critical:	
1	Base assembly seal ring missing or wrong type
Major:	AQL 1.0 percent defective
101	Component missing or assembled incorrectly
102	Cover depth from ogive face incorrect
103	Base torque incorrect
104	Marking illegible or incorrect
105	Zone marking missing, incorrect or unidentifiable
106	Lifting plug missing or wrong type (not fusible type)
107	Base assembly seal ring not greased
108	Workmanship (3.5)

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(b) Canister spacer (Drawing D15-12-167).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Overall diameter incorrect
102	Overall length incorrect
103	Assembly incorrect
104	Workmanship (3.5)
Minor:	AQL 2.5 percent defective
201	Materials incorrect
202	Finish incorrect
203	Marking missing or incorrect

(c) Chemical safe indicator (Drawing C15-12-166).

Categories	Defects
Critical:	None defined
Major:	None defined
Minor:	AQL 2.5 percent defective
201	Material incorrect
202	Inside diameters incorrect
203	Material thickness incorrect
204	Overall height incorrect
205	Radii incorrect
206	Marking missing or incorrect
207	Height to inside lip incorrect
208	Workmanship (flash, pinholes, 3.5)

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(d) Wrench (Drawing D15-12-185).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Materials incorrect
102	Flat bar length incorrect
103	Flat bar radii incorrect
104	Flat bar weldments incorrect
105	Wrench opening incorrect
106	Workmanship, chamfers missing (3.5)
Minor:	AQL 2.5 percent defective
201	Tubing length incorrect
202	Drive screws improperly installed
203	Finish incorrect
204	Round bar length incorrect
205	Round bar position incorrect

(e) Thickness gauge (Drawing C15-12-221).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Thickness incorrect
102	Packaging incorrect
Minor:	AQL 2.5 percent defective
201	Materials incorrect
202	Marking incorrect

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TABLE I. Ballistic functioning

Test	Propellant: Charge	Zone	Elevation: mils	Prepro quantity	Production: quantity	Requirement paragraph
Canister functioning	M1	1	300**	4	2	3.4.1 3.4.2
Projectile functioning	M118E1	9	300**	4	2	3.4.2
Projectile functioning	M118E1	9+ (15%)	300**	3	1	3.4.2
Replacement sample				1	1	*

*NOTE: Retest: If for any reason the proving ground considers that test conditions have affected all or part of the test results, the proving ground shall declare the affected results invalid and perform new tests on additional projectiles, as required. If retest is required, replacement sample shall be tested at zone 9 plus 115% and test results recorded.

**Subject to change with Chemical Systems Laboratory, ARRADCOM approval.

6. NOTES

6.1 Intended use. The projectile will provide the armed forces with a lethal agent retaliatory capability, but will not present a lethal hazard during storage and handling.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Federal stock number or other identifying number.
(c) Nomenclature of item.
(d) Destination address.
(e) Transportation control number.
(f) Transportation priority.
(g) Any special marking.

6.3 Simulant. Simulant canisters (Drawing D15-12-213) shall be TEP/DBA/EA. WARNING. Use of QL, XM27 canister will produce lethal VX agent.

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6.4 Supersession data. This specification supersedes Chemical Systems Laboratory Purchase Description EA-P-1170, 31 May 1977.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1320-A472

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